

Scientific Words of the Day

30 root puzzles from physics to environmental science

Physics first, Earth systems next

prefix + root + suffix → physical model

How to use one pair of slides each day

1. Decide whether the prompt asks for a root or a meaning.
2. Choose A or B and justify the choice.
3. Use the cloze sentence to infer the complete word.
4. Sketch the process or system boundary.
5. Turn the page and test both answers.

Recurring questions

What moves? In what form? Across which boundary? What changes? What remains unchanged?

Root key

L = Latin origin

G = Greek origin

Teacher note

The root gloss is a memory prompt. The scientific context supplies the precise modern meaning. Many science words combine Greek and Latin elements.

LIGHT, RADIATION AND TRANSFER

ROOT TO MEANING

Which meaning belongs to this root?

flect L

A

bend

B

break

Complete the sentence Light undergoes _____ when it returns from a mirror or water surface.

Choose, then infer the scientific word.
Justify both answers before turning the page.

reflection

Root image: the process of bending back

In context At a mirror or water surface, light returns to the original medium.

Physics

For a plane mirror, the angles are measured from the normal: $\theta_i = \theta_r$.

Environmental connection

Ice and clouds reflect sunlight. Their reflectivity contributes to Earth's albedo.

Word family: reflect, reflection, reflector, reflective

Check: What changes in reflection: the wave speed, its direction, or both?

LIGHT, RADIATION AND TRANSFER

MEANING TO ROOT

Which root carries this meaning?

break

A

fract **L**

B

flect **L**

Complete the sentence Atmospheric _____ can make the Sun visible just before geometric sunrise.

Choose, then infer the scientific word.
Justify both answers before turning the page.

refraction

Root image: a bending or “breaking” of the path

In context Layers of air bend the apparent path of sunlight near the horizon.

Physics

The refractive index is $n = c/v$. A change in wave speed produces a change in direction.

Environmental connection

Layers of warm and cool air refract light differently, producing mirages.

Word family: refract, refraction, refractive, refractor

Check: Why does a ray entering along the normal change speed without changing direction?

LIGHT, RADIATION AND TRANSFER

ROOT TO MEANING

Which meaning belongs to this root?

miss L

A

send

B

carry

Complete the sentence Glass allows high _____ of visible light.

Choose, then infer the scientific word.

Justify both answers before turning the page.

transmission

Root image: the process of sending across

In context Visible light can pass through clear glass.

Physics

A transmission fraction can be written

$$T = I_{\text{trans}} / I_{\text{incident}}.$$

Environmental connection

The atmosphere has transmission windows through which visible light and some radio waves pass.

Word family: transmit, transmission, transmitter, transmissive

Check: If 70% is transmitted, what fraction remains for reflection and absorption?

LIGHT, RADIATION AND TRANSFER

MEANING TO ROOT

Which root carries this meaning?

swallow up

A

sorpt **L**

B

vect **L**

Complete the sentence Dark surfaces show strong _____ of visible radiation.

Choose, then infer the scientific word.
Justify both answers before turning the page.

absorption

Root image: the process of swallowing up

In context A dark surface takes radiant energy into its internal energy store.

Physics

Absorbed radiation transfers energy to matter, often raising its internal energy.

Environmental connection

Greenhouse gases absorb selected infrared wavelengths; oceans absorb most of Earth's excess heat.

Word family: absorb, absorption, absorptive, absorber

Check: How is absorption different from a wave merely disappearing from view?

Which meaning belongs to this root?

e- / ex- L

A

out

B

back

Complete the sentence Every warm body produces thermal _____.

Choose, then infer the scientific word.

Justify both answers before turning the page.

emission

Root image: the process of sending out

In context A warm body sends thermal radiation into its surroundings.

Physics

A grey body's radiated power is $P = \epsilon \sigma AT^4$.

Environmental connection

Earth emits infrared radiation; vehicles also emit gases and particles. The same word can name energy or matter leaving a system.

Word family: emit, emission, emitter, emissivity

Check: In each use of “emission”, what crosses the system boundary?

LIGHT, RADIATION AND TRANSFER

MEANING TO ROOT

Which root carries this meaning?

lead

A

duct **L**

B

vect **L**

Complete the sentence A metal spoon warms by thermal _____.

Choose, then infer the scientific word.

Justify both answers before turning the page.

conduction

Root image: the process of leading together

In context Energy passes along a metal spoon through neighbouring particle interactions.

Physics

For a simple slab, the transfer rate is $P = kA\Delta T/L$.

Environmental connection

Conduction through walls affects building energy demand; soil conduction affects permafrost thaw.

Word family: conduct, conduction, conductor, conductivity

Check: Which material property in the equation measures how readily energy is conducted?

Which meaning belongs to this root?

vect L

A

carry

B

bend

Complete the sentence Warm air rises and cooler air replaces it in a _____ current.

Choose, then infer the scientific word.
Justify both answers before turning the page.

convection

Root image: the process of carrying together

In context Rising warm air carries internal energy while cooler air replaces it.

Physics

Convection combines bulk fluid motion with energy transfer. It requires a liquid or gas.

Environmental connection

Atmospheric and oceanic convection redistribute heat, moisture and dissolved substances.

Word family: convect, convection, convective; compare advection

Check: What is transported in convection besides energy?

LIGHT, RADIATION AND TRANSFER

MEANING TO ROOT

Which root carries this meaning?

ray

A

radi **L**

B

insul(a) **L**

Complete the sentence The Sun transfers energy through space by electromagnetic

_____.

Choose, then infer the scientific word.

Justify both answers before turning the page.

radiation

Root image: energy travelling as rays

In context Energy from the Sun crosses space as electromagnetic waves.

Physics

Electromagnetic waves in vacuum obey $c = f\lambda$.

Environmental connection

Incoming solar radiation and outgoing infrared radiation form Earth's planetary energy budget.

Word family: radiate, radiation, radiator, irradiance

Check: Why can radiation cross space while conduction and convection cannot?

LIGHT, RADIATION AND TRANSFER

ROOT TO MEANING

Which meaning belongs to this root?

insul(a) L

A

island

B

balance

Complete the sentence Foam provides thermal _____ by reducing energy transfer.

Choose, then infer the scientific word.

Justify both answers before turning the page.

insulation

Root image: the process of making into an island

In context Foam reduces thermal transfer between a region and its surroundings.

Physics

Insulation lowers the transfer rate; it does not prevent energy transfer perfectly.

Environmental connection

Building insulation reduces fuel demand. Snow also insulates soil and organisms beneath it.

Word family: insulate, insulation, insulator, insulating

Check: Which transfer mechanisms can a layer of insulation reduce?

LIGHT, RADIATION AND TRANSFER

MEANING TO ROOT

Which root carries this meaning?

between

A

inter **L**

B

con **L**

Complete the sentence A force describes an _____ between bodies.

Choose, then infer the scientific word.

Justify both answers before turning the page.

interaction

Root image: an action between things

In context A force exists between two bodies and affects both.

Physics

Interactions occur in pairs: if A exerts a force on B, B exerts an equal and opposite force on A.

Environmental connection

Environmental systems contain linked physical, chemical and biological interactions.

Word family: interact, interaction, interactive, actor

Check: Why is a force better represented between two bodies than as a possession of one body?

Which meaning belongs to this root?

celer L

A

swift

B

heavy

Complete the sentence A falling raindrop undergoes downward _____ until drag balances its weight.

Choose, then infer the scientific word.

Justify both answers before turning the page.

acceleration

Root image: the process of becoming swifter

In context A falling raindrop changes its velocity downward until drag balances its weight.

Physics

Average acceleration is $a = \Delta v / \Delta t$. Direction may change even when speed does not.

Environmental connection

Wind acceleration and changing currents influence the transport of heat, moisture and pollutants.

Word family: accelerate, acceleration, accelerator

Check: Can an object accelerate while moving at constant speed? Give an example.

MOTION, FORCES AND SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

heavy

A

grav **L**

B

celer **L**

Complete the sentence Newton's law of universal _____ describes the attraction between masses.

Choose, then infer the scientific word.

Justify both answers before turning the page.

gravitation

Root image: the process associated with heaviness

In context Every mass attracts every other mass across the space between them.

Physics

For two point masses, $F = GMm/r^2$. The field is an interaction with mass.

Environmental connection

Gravity drives sediment settling, river flow, atmospheric pressure gradients and ocean tides.

Word family: gravity, gravitation, gravitational, gravimeter

Check: Which variable in the inverse-square law describes separation rather than either object?

MOTION, FORCES AND SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

tract L

A

draw or pull

B

press

Complete the sentence Unlike electric charges experience electrostatic _____.

Choose, then infer the scientific word.

Justify both answers before turning the page.

attraction

Root image: the process of drawing towards

In context Unlike electric charges pull towards one another.

Physics

Attraction describes the direction of the force; it does not identify the interaction responsible.

Environmental connection

Attractive forces help aerosol particles cluster and affect how contaminants attach to surfaces.

Word family: attract, attraction, attractive, tractor

Check: What additional information is needed before “attraction” becomes a physical explanation?

MOTION, FORCES AND SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

press

A

press **L**

B

pans **L**

Complete the sentence Rapid _____ of a gas usually raises its temperature.

Choose, then infer the scientific word.

Justify both answers before turning the page.

compression

Root image: the process of pressing together

In context Pressing a gas into a smaller volume usually raises its temperature.

Physics

Work done on a gas transfers energy into it. At fixed temperature, pV is approximately constant for an ideal gas.

Environmental connection

Descending air is compressed and warms, influencing cloud formation and local weather.

Word family: compress, compression, compressor, compressible

Check: Why does compression require an energy-transfer account as well as a volume account?

MOTION, FORCES AND SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

pan L

A

spread

B

keep

Complete the sentence A gas cools during adiabatic _____ as it does work on its surroundings.

Choose, then infer the scientific word.
Justify both answers before turning the page.

expansion

Root image: the process of spreading out

In context A gas spreads into a greater volume and cools as it does work on its surroundings.

Physics

Thermal expansion also changes the dimensions of solids and liquids as temperature changes.

Environmental connection

Rising air expands and cools; water vapour may then condense to form clouds.

Word family: expand, expansion, expansive, expansivity

Check: What system does the expanding gas do work on?

MOTION, FORCES AND SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

swing

A

oscill **L**

B

libr **L**

Complete the sentence A pendulum's displacement undergoes an _____ about equilibrium.

Choose, then infer the scientific word.

Justify both answers before turning the page.

oscillation

Root image: the process of swinging

In context A pendulum repeatedly moves about its equilibrium position.

Physics

Period and frequency are reciprocal: $T = 1/f$.
Energy shifts between stores during each cycle.

Environmental connection

Ocean-atmosphere patterns such as El Niño involve large-scale, irregular oscillations.

Word family: oscillate, oscillation, oscillator, oscillatory

Check: What defines the equilibrium value about which a system oscillates?

MOTION, FORCES AND SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

libr L

A

balance

B

shape

Complete the sentence At thermal _____, bodies in contact have no net energy transfer between them.

Choose, then infer the scientific word.

Justify both answers before turning the page.

equilibrium

Root image: a state of equal balance

In context Bodies in thermal contact reach equal temperature, giving zero net energy transfer.

Physics

Equilibrium is often dynamic: microscopic exchanges continue while opposing rates balance.

Environmental connection

Planetary radiative equilibrium requires average incoming and outgoing energy fluxes to be equal.

Word family: equilibrate, equilibrium, equilibrant

Check: Why does equilibrium mean zero net change rather than zero activity?

MOTION, FORCES AND SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

keep

A

serv **L**

B

sipat **L**

Complete the sentence Energy _____ requires the total energy of a closed system to remain constant.

Choose, then infer the scientific word.

Justify both answers before turning the page.

conservation

Root image: the process of keeping together

In context In an isolated system, total energy remains constant while being redistributed.

Physics

A conservation law is a bookkeeping constraint:

$\Delta E_{\text{total}} = 0$ for an isolated system.

Environmental connection

Environmental conservation means protection and careful management. “Conserve energy” concerns use and quality, since the amount itself is conserved.

Word family: conserve, conservation, conservative, conserved

Check: How do the physics and environmental meanings of conservation differ?

MOTION, FORCES AND SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

sipat L

A

scatter

B

send

Complete the sentence Friction causes _____ of mechanical energy into thermal energy stores.

Choose, then infer the scientific word.

Justify both answers before turning the page.

dissipation

Root image: the process of scattering apart

In context Friction spreads organised mechanical energy into thermal stores.

Physics

Energy remains conserved while becoming more dispersed and less available for useful work.

Environmental connection

Engines, electrical systems and cities dissipate energy as waste heat; efficiency limits fuel demand and environmental impact.

Word family: dissipate, dissipation, dissipative

Check: How can energy be both conserved and dissipated?

MOTION, FORCES AND SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

shape

A

form **L**

B

grad **L**

Complete the sentence A motor produces an energy _____ from electrical input to kinetic and thermal stores.

Choose, then infer the scientific word.

Justify both answers before turning the page.

transformation

Root image: the process of changing shape or form

In context A motor changes electrical input into kinetic and thermal energy stores.

Physics

A transformation changes representation or form; a transfer crosses a system boundary.

Environmental connection

Photosynthesis transforms radiant energy into chemical energy stored in biomass.

Word family: transform, transformation, transformer, transformative

Check: Is heating a transformation, a transfer, or potentially both? Define the system first.

ENVIRONMENTAL SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

pollut L

A

defile or soil

B

nourish

Complete the sentence Air _____ includes harmful gases and suspended particles.

Choose, then infer the scientific word.

Justify both answers before turning the page.

pollution

Root image: the result of making an environment unclean

In context Harmful gases and suspended particles can enter the air.

Physics

Physics describes sources, diffusion, flow, settling and exposure; harm also depends on chemistry and biology.

Environmental connection

Pollution may involve matter, noise, light or thermal energy. It is defined partly by its damaging effects.

Word family: pollute, pollution, pollutant, polluting

Check: Why is the presence of a substance alone insufficient to establish pollution?

ENVIRONMENTAL SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

step

A

grad **L**

B

pans **L**

Complete the sentence Energy _____ describes the loss of its capacity to produce useful work.

Choose, then infer the scientific word.

Justify both answers before turning the page.

degradation

Root image: the process of stepping down

In context Dissipation lowers energy's capacity to produce useful work.

Physics

The total energy remains conserved while dissipation lowers its usefulness.

Environmental connection

Land degradation reduces soil structure, fertility, biodiversity or productive capacity.

Word family: degrade, degradation, degradable, biodegradable

Check: Which measurable indicators would show that an ecosystem has degraded?

ENVIRONMENTAL SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

atmos G

A

vapour

B

life

Complete the sentence Earth's _____ is a compressible fluid held by gravity.

Choose, then infer the scientific word.

Justify both answers before turning the page.

atmosphere

Root image: the globe or region of vapour

In context A compressible fluid of gases surrounds Earth and is held by gravity.

Physics

Pressure, density, convection, radiation and phase changes govern its behaviour.

Environmental connection

The atmosphere transports energy and water, shields life and exchanges matter with oceans, land and organisms.

Word family: atmosphere, atmospheric; compare hydrosphere and biosphere

Check: Why is the atmosphere best modelled as both a layer and a dynamic system?

ENVIRONMENTAL SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

human

A

anthropo **G**

B

bio **G**

Complete the sentence The observed rise in atmospheric carbon dioxide has a large _____ component.

Choose, then infer the scientific word.
Justify both answers before turning the page.

anthropogenic

Root image: produced by humans

In context Human activity contributes substantially to the observed rise in atmospheric carbon dioxide.

Physics

A causal claim requires a mechanism, measurements and comparison with natural drivers.

Environmental connection

Anthropogenic forcing includes greenhouse-gas emissions, aerosols and land-use change.

Word family: anthropogenic, anthropogenicity; compare biogenic and geogenic

Check: Why does “anthropogenic” identify an origin rather than automatically assign blame?

ENVIRONMENTAL SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

divers L

A

different

B

great

Complete the sentence A coral reef may contain high _____ across several levels of organisation.

Choose, then infer the scientific word.

Justify both answers before turning the page.

biodiversity

Root image: the state of varied life

In context A coral reef can contain wide variation among genes, species and ecosystems.

Physics

Biodiversity is measured through richness, evenness and related indices; no single number captures every aspect.

Environmental connection

Diverse systems may possess more response pathways when conditions change, though diversity alone does not guarantee resilience.

Word family: biology, biological, diversity, biodiverse

Check: What information is lost if biodiversity is reported only as a species count?

ENVIRONMENTAL SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

heap up

A

accumul **L**

B

sequestr **L**

Complete the sentence Mercury can undergo _____ within an individual organism over time.

Choose, then infer the scientific word.

Justify both answers before turning the page.

bioaccumulation

Root image: the process of heaping up in life

In context Mercury can build up within an individual organism over time.

Physics

A mass balance compares input rate with metabolism and excretion. Accumulation occurs when net input is positive.

Environmental connection

Persistent, fat-soluble substances are especially likely to accumulate in tissues.

Word family: accumulate, accumulation, bioaccumulate, bioaccumulation

Check: What two rates must be compared to predict whether concentration will rise?

ENVIRONMENTAL SYSTEMS

ROOT TO MEANING

Which meaning belongs to this root?

magn L

A

great

B

equal

Complete the sentence A persistent pesticide may show _____ from prey to predator.

Choose, then infer the scientific word.
Justify both answers before turning the page.

biomagnification

Root image: the process of making greater through life

In context Persistent pesticide concentrations can increase from prey to predator.

Physics

A predator consumes contaminant stored in many prey organisms, producing a trophic concentration increase.

Environmental connection

Top predators may receive the largest dose even when environmental concentrations are low.

Word family: magnify, magnification, biomagnify, biomagnification

Check: How does biomagnification differ from bioaccumulation?

ENVIRONMENTAL SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

set apart

A

sequestr **L**

B

serv **L**

Complete the sentence Carbon _____ stores carbon in biomass, soils, oceans or geological formations.

Choose, then infer the scientific word.

Justify both answers before turning the page.

sequestration

Root image: the process of setting apart

In context Carbon can be stored in biomass, soils, oceans or geological formations.

Physics

A useful account specifies the reservoir, transfer rate, storage capacity and residence time.

Environmental connection

Storage contributes to mitigation only while carbon remains isolated from the atmosphere for a relevant timescale.

Word family: sequester, sequestration, sequestered

Check: Why is the amount stored insufficient without a timescale?

Which meaning belongs to this root?

troph G

A

nourishment

B

vapour

Complete the sentence Fertiliser runoff can cause lake _____.

Choose, then infer the scientific word.

Justify both answers before turning the page.

eutrophication

Root image: the process of becoming richly nourished

In context Fertiliser runoff supplies excess nutrients and can eventually lower dissolved oxygen.

Physics

Nutrients increase algal growth; decomposition then raises biological oxygen demand and lowers dissolved oxygen.

Environmental connection

A resource that initially increases production can destabilise the wider aquatic system.

Word family: trophic, eutrophic, eutrophication, oligotrophic

Check: Why can greater nutrient supply ultimately reduce animal life in the water?

ENVIRONMENTAL SYSTEMS

MEANING TO ROOT

Which root carries this meaning?

hold

A

tain **L**

B

tract **L**

Complete the sentence Environmental _____ asks whether a practice can persist within physical and ecological limits.

Choose, then infer the scientific word.

Justify both answers before turning the page.

sustainability

Root image: the capacity to keep holding up or enduring

In context A practice can persist only while its physical and ecological supports remain available.

Physics

The question requires stocks, flows, efficiencies, feedbacks, boundaries and timescales.

Environmental connection

A process is sustainable only relative to stated environmental, social and temporal conditions.

Word family: sustain, sustainable, sustainability, sustained

Check: What would have to remain available or regenerate for this system to continue?

From word to world

root $\longrightarrow$ action $\longrightarrow$ model $\longrightarrow$ system

What moves? In what form? Across which boundary?

What changes? What accumulates? What remains unchanged?
